

Kartikeya Bhatnagar

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Education

Bennett University (Times of India Group)

B.Tech in Computer Science and Engineering (AI Specialization) | CGPA: 9.33/10

Sep 2022 – Present

Greater Noida, UP

Kendriya Vidyalaya No.2, AFS Hindon (CBSE)

Senior Secondary School (XII), Computer Science | Score: 96.2%

Aug 2012 – May 2021

Ghaziabad, UP

Research & Publications

IEEE Conference Paper

Feb 2025

Performance Comparison of Deep Learning Models in Predicting Stock Prices

- Authored a peer-reviewed comparative study of four architectures (LSTM, GRU, CNN1D, SimpleRNN) for 1-day-ahead stock prediction across 20 Indian companies in 5 sectors.
- Benchmarked models across MAE, RMSE, MAPE, NRMSE, and R^2 ; GRU delivered top accuracy (MAE 16.50, RMSE 22.59, $R^2 \approx 0.99$).
- DOI: <https://ieeexplore.ieee.org/document/10932578>

Experience

Airtel (Bharti Airtel Ltd.)

Jun 2025 – Aug 2025

AI Intern

Gurgaon, Haryana

- Engineered a geospatial pipeline to construct and refine urban building-cluster polygons using GeoPandas, Shapely, SciPy, spatial indexing (KD-tree/Rtree), and CRS-aware transforms.
- Architected a two-stage workflow: (1) extracted dense polygons via Delaunay triangulation with adaptive circumradius thresholds; (2) aligned boundaries to roads with outward-only snapping and spatial indexing.
- Verified 99%+ topological correctness post-smoothing; constrained area change to 2–5% and reduced boundary-to-road distance for cartographic alignment.
- Maintained mean vertex displacement ≤ 40 m to preserve shape integrity and ensure clean integration with transportation networks.

Projects

Sketch 2 Image (GitHub)

Sep 2025 – Nov 2025

- Implemented real-time sketch-to-image generation with OpenCV (webcam), Stable Diffusion (Hugging Face), and Gemini-driven prompts.
- Processed 10,000+ hand-tracking gestures per session using MediaPipe.

Vulgar Veto (GitHub)

Jan 2025 – Mar 2025

- Developed a Streamlit/Python application for speech transcription and offensive-term filtering using a curated lexicon of 1,000+ terms.
- Increased accuracy by segmenting audio into small windows; achieved $\sim 90\%$ transcription success (Google Speech Recognition) and produced filtered playback via gTTS.

Stock Prediction using Deep Learning (GitHub)

Jan 2024 – May 2024

- Benchmarked LSTM, GRU, CNN, and RNN on MSFT and Nifty50 datasets; achieved $R^2 = 0.994$ for Technology stocks using GRU.
- Analyzed 10+ years of market data; reduced RMSE by 15% through hyperparameter tuning and sequence-window optimization.

Technical Skills

Programming Languages: Python, C++, Java, SQL, HTML, CSS

Machine Learning / AI: Supervised & Unsupervised ML, Time Series, Deep Learning (CNN, RNN, LSTM, GRU), NLP, Computer Vision

Databases: SQLite3, MySQL

Libraries & Frameworks: Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, Keras, TensorFlow, Django, MediaPipe, OpenCV, Streamlit, gTTS, GeoPandas, Shapely, Folium, Rtree

Certifications

- Exploratory Data Analysis for Machine Learning – IBM (certificate)
- Supervised Machine Learning: Classification – IBM (certificate)
- Unsupervised Machine Learning – IBM (certificate)
- Convolutional Neural Networks – DeepLearning.AI (certificate)
- Natural Language Processing – DeepLearning.AI (certificate)
- Introduction to Modern Database Systems – Saylor.org (certificate)
- Data Structures – University of California San Diego (certificate)
- Algorithmic Toolbox – University of California San Diego (certificate)